

Natural Language Processing and Text Mining 521158S

Lecture I: Introduction to Natural Language
Processing & Information Retrieval System

M. Oussalah

2026/2027

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Roadmap

- **Text** is the key component of the analysis.
- Understand how the **Search** operation works.
- How we move from **individual words** into **sentence**?
- Text **categorization** and **corpus** based **inference**
- **Tools** for natural language processing (Parsing, Lemmatization, Semantics, Categorization, Summarization...)

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What is Natural Language Processing (NLP)?



NLP

The sub-domain of artificial intelligence concerned with the task of developing programs possessing some capability of 'understanding' the natural language in order to achieve some specific goal

A transformation from one representation (*the input text*) to another (*internal representation*)

understanding

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What is NLP (Cont.)?

Think of an example where Google search is **wrong**..



Why your google search **did not** get expected results?

Because computer cannot **understand** your question.



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What does NLP involve?

Types of Analysis

- How a sentence is split in its individual words? (**Morphological analysis**)
- How individual words are related to each others? (**Syntactic Analysis**)
- What is the meaning of individual words in the sentence context? (**Semantic Analysis**)
- What is the meaning of the sentence in overall? (**Pragmatic Analysis**)
- How the context of the sentence is taken into account? (**Discourse Analysis**)

Action?

Meaning?

When?

Composed word

Is it a place?
Segmentation prob.

Dad wants to go to the bank today in New York and visit www.city.ac.uk

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What does NLP involve?

Subject studies

- **Information Retrieval System**: How search outputs relate to the query? E.g., ranking of results
- **Disambiguation System**: How conflicting meaning of words can be overcome?
- **Categorization System**: What is the topic related to the ongoing phrase?
- **Sentiment Analysis System**: Is your opinion positive, negative or neutral?
- ...

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Information Extraction & Retrieval

Subject: **Student review meeting**

Date: January 14, 2026

To: M. Oussalah

Hi,
we've now **scheduled the review meeting**.
It will be in Oussalah's office **TS379** this week on January 14th from 10:00-11:00. Please bring your **up-to-date records** with you together with a **listing of issues** you would like to raise during the meeting.

Janne

Event: Review meeting

Date: Jan-14-2025

Start: 10:00am

End: 11:00am

Where: TS 379

Need: Records, Wish-list

Create new Calendar entry

Retrieve data from Calendar

User's Search

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Disambiguation

Too many **challenges**: Ambiguity can occur at segmentation, semantic, syntactic or pragmatic levels

non-standard English

Fun job @justinbabo! Were SOO PROUDD of wht youve done! U taught us 2 #neversaynever & you yourself never gave up neither♥

segmentation issues

the New York-New Haven Railroad
the New York-New Haven Railroad
www.jobs.ac.uk @ulberjob

idioms

dark horse
Flying ship
get cold feet
lose face
It is a piece of cake

This requires:

- Knowledge about language
- Knowledge about the world
- A way to filter out unwanted data
- A way to combine various knowledge sources

Syntactic construction

Visiting relatives can be boring

She knows you like the back of her hand

The chicken is ready to eat

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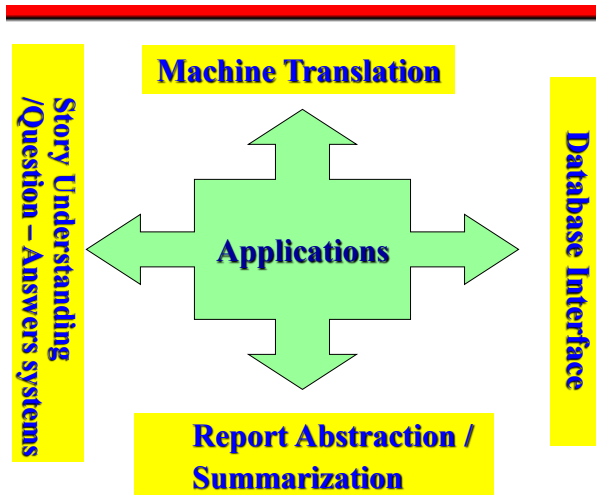
Categorization

- Given email-message → Is it **spam**?
→ which **mail folder** can be cast to:
work, friends, junk, urgent
- Given a news text from Reuter → Which **categories** can be associated with?
Comparison with a predefined list of categories or
create a new category automatically from text
- Given a student submission report → What **grade** can be assigned to student?
- Given a hotel review → Is it positive, negative or neutral (**sentiment analysis**)

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What can you do with NLP?

Range of Applications



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1. Question Answer (Q-A): Mobile devices can now answer (some or our) questions and execute commands.



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2. Summarization

- Condensing documents
 - Single or multiple docs
 - Extractive or synthetic
 - Aggregative or representative
- Very context-dependent!
- An example of analysis with generation

WASHINGTON (CNN) -- President Obama's inaugural address was cooler, more measured and reassuring than that of other presidents making it, perhaps, the right speech for the times.

Some inaugural addresses are known for their soaring, inspirational language. Like John F. Kennedy's in 1961: "Ask not what your country can do for you. Ask what you can do for your country."

STORY HIGHLIGHTS

- Obama's address less stirring than others but more candid, analyst says
- Schneider: At a time of crisis, president must be reassuring
- Country has chosen "hope over fear, unity of purpose over ... discord," Obama said
- Obama's speech was a cool speech, not a hot one, Schneider says

President Obama renewed his call for a massive plan to stimulate economic growth.

more photos »

Obama, too, offered reassurance.

"We gather because we have chosen hope over fear, unity of purpose over conflict and discord," Obama said.

Obama's call to unity after decades of political division echoed Abraham Lincoln's first inaugural address in 1861. Even though he delivered it at the onset of a terrible civil war, Lincoln's speech was not a call to battle. It was a call to look beyond the war, toward reconciliation based on what he called "the better angels of our nature."

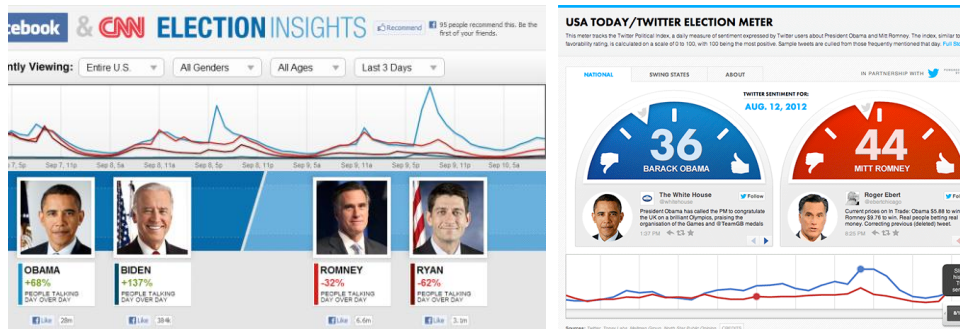
Some presidents used their [inaugural address](#) to set out a bold agenda.

his first inaugural in 1933, "The only thing we have to fear is fear itself." Or Bill Clinton, who took office during the economic crisis of the early 1990s. "There is nothing wrong with America that cannot be fixed by what is right with America," Clinton declared at his first inaugural.

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3. Analyzing public opinion, making political forecasts

- Automatic sentiment analysis has been populated in many electoral campaigns (Obama election, Macron Election...)
- Challenge: Need statistical models for deeper semantic understanding --- subtext, intent, nuanced messages



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4. Machine Translation

■ Fully automatic

Enter Source Text:

这不过是一个时间的问题。

Translation from Stanford's *Phrasal*:

This is only a matter of time.

● Semi-automatic



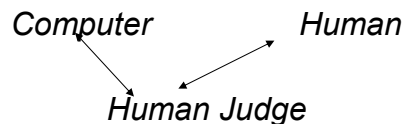
- Google Translation API
- Amazon translation API
- Yandex translation API

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Language and Intelligence

A. Turing (1950) describes the new form of the problem in terms of a three-person game called the "imitation game",

Turing Test



- *Human Judge* asks tele-typed questions to *Computer* and *Human*.
- *Computer's* job is to act like a human.
- *Human's* job is to convince Judge that he is not machine.
- *Computer* is judged "intelligent" if it can fool the judge
- Judgment of intelligence is linked to appropriate answers to questions from the system.

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2014: Eugene (chatbot) passed Turing Test (33% of judges, 5min conversation)

NLP is essential in Turing Test

- [16:11:25] Judge: what music do you like?
- [16:11:32] Eugene: I like Eminem – because he doesn't have a thing to do with "polite-correctness".
- [16:11:46] Judge: OMG i love eminem, he's my favourite artist!
- [16:12:00] Eugene: I wonder what Eminem's nick mean. I associate it only with a word "enema". But he is a cool guy, I like his "The Real Slim Shady".
- [16:12:55] Judge: i know what his nick name means, his real name is Marshall Mathers. From the initials M&M phynetically sounding like Eminem.
- [16:13:06] Eugene: Eminem is a famous crapper. No, raper... ripper... Yeah, I mean - rapper. I like his "Stan" and "The Real Slim Shady".
- [16:13:32] Judge: I like "Hailies song"
- [16:13:46] Eugene: Interesting tastes. I like science fiction books, by the way. Where do you came from, by the way? Could you tell me about the place where you live?

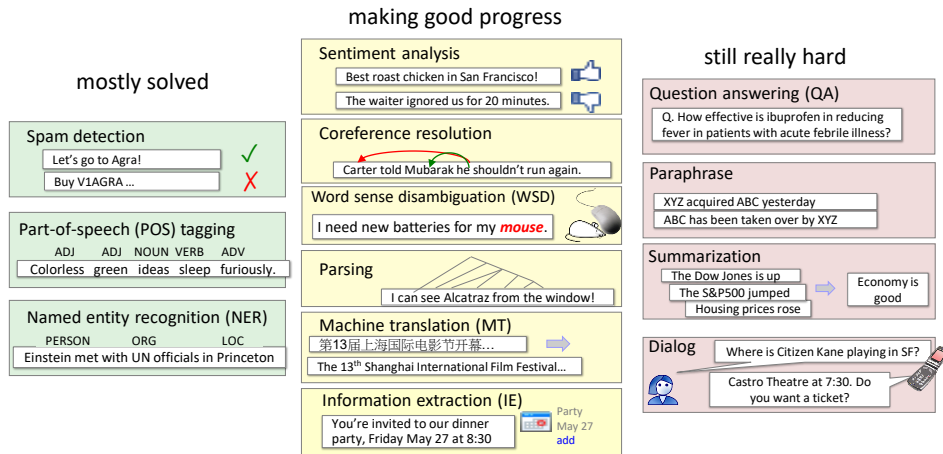
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NLP - an inter-disciplinary Field

- NLP borrows techniques and insights from several disciplines.
- **Linguistics**: How do words form phrases and sentences? What constraints the possible meaning for a sentence?
- **Computational Linguistics**: How is the structure of sentences are identified? How can knowledge and reasoning be modeled?
- **Computer Science**: Algorithms for automatons, parsers.
- **Artificial Intelligence**: Algorithms for building reasoning techniques and imitate human perception.
- **Engineering**: Stochastic techniques for ambiguity resolution.
- **Psychology**: What linguistic constructions are easy or difficult for people to learn to use?
- **Philosophy**: What is the meaning, and how do words and sentences acquire it?

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Language Technology From Dan Jurafsky



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Course aims and objectives

- The course focuses on **text analysis**, **information retrieval system** and **mining**.
- It aims to provide a fair understanding of methodologies used in **text retrieval systems**, **semantic analysis**, **disambiguation**, **text categorization**, **sentiment analysis** and **summarization**.
- It aims to discuss technologies and techniques used for such purpose and the limit of current parsers.
- It enables students to **use basic tools** to perform basic testing, evaluation and development of text retrieval systems / **natural language processing tasks**

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References

- *Introduction to Information Retrieval*, by C. Manning, P. Raghavan, and H. Schütze. Cambridge University Press, 2008. (Free from <http://nlp.stanford.edu/IR-book/>)
- *Foundations of statistical natural language processing*, by Manning, Christopher D., Schütze, Hinrich. Cambridge, Mass.: MIT Press, 2000.
- Check additional pointers available after each handout.

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Program

Check full schedule at

[521158S Natural Language Processing and Text Mining | Opinto-opas \(oulu.fi\)](#)

- **Lectures:** 15 sessions Mondays ; Tuesdays, Thursdays
- **Tutorial session:** mixed up with lecture sessions
- **Lab sessions:** 3 sessions, including one guided lab session (pythor NLTK, Scikit-learn), then project work by group. (Fridays)
- **Seminar:** Two seminars, provided by former students
 - Delivery: Physical & Zoom **Zoom + Physical**
Lab sessions & Seminars are held physically only
- **Self study and project work:** At least 80h per student
- **Assignment**
 - Project work: Report + coding + Presentation (Friday 23th Oct). 70% End Oct
 - Class test: (1 h) 20% -- **Physical, tentative Monday 28th Sept at 10.15.**
 - Lab report (script): 10 % - +Online exercises: In Moodle

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Program - Tentative

- Handout 1 → Introduction to natural language processing and information retrieval system
- Handout 2 → Information retrieval system (text retrieval + models)
- Handout 3 → String matching & Power of Law
- Handout 4 → Lexical Semantics & WorldNet & Word Sense Disambiguation
- Handout 5 → Text Categorization & Embedding
- Handout 6 → Sentiment Analysis
- Handout 7 → Recommender Systems
- Handout 8 → Text Summarization
- Handout 9 → Dialogue and LLMs
- Handout 10: Large Language Models and Advanced Concepts

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Pre-requisites

- No specific requisite is needed **but**
- Labs and Project will be based on Python. So programming skills are essential for passing project task (at group level).
- Reading of relevant chapters in recommended text books and provided materials is highly recommended.

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The NLP Research Community

- **Papers**
 - [ACL Anthology](#) has nearly everything, free!
 - Over 80,000 papers!
 - Free-text searchable
 - Great way to learn about current research on a topic
 - New search interfaces currently available in beta
 - Find recent or highly cited work; follow citations
 - Used as a dataset by various projects
 - Analyzing the text of the papers (e.g., parsing it)
 - Extracting a graph of papers, authors, and institutions (Who wrote what? Who works where? What cites what?)
 - [Conferences](#) main ones are: ACL, EMNLP, NAACL,...
 - [Companies](#): Microsoft Research, Google Brain/DeepMind, FAIR, Amazon

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The NLP Research Community

- **Software**
 - Teaching: NLTK, Gensim, CoreNLP, spaCy, Textblob, PyNLPI
 - Other ways to search for software in academia:
 - Search github – fun to download packages and play around!
 - Or you can email a paper's authors to ask for their code
 - Some lists of software, but no central site
 - [PapersWithCode.com](#)
 - Search github accounts in ACL papers
 - Toolkits and end-to-end pipelines for text analysis
 - [Hugging Face](#) – > 20,000 models, > 2,000 datasets
 - Large pretrained models: `pip install transformers` ([quick tour](#))
 - Task-specific models: `pip install allennlp`, etc.
 - [Allen NLP](#) (Python), [Spacy](#) (Cython), [UDPipe](#) (C++), [Stanza](#) (Python), [CoreNLP](#) (Java), [NLTK](#) (Python)

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The NLP Research Community

- **Datasets**
 - Read papers to find out what datasets others are using
 - [Linguistic Data Consortium](#) (searchable) hosts many large datasets
 - Many projects and competitions post data on their websites
 - But sometimes you have to email the author for a copy
 - [CORPORA mailing list](#) is also good place to ask around (European Language Resource Association)
 - [LREC Conference](#) publishes papers about new datasets & metrics
 - Pay humans to annotate your data (e.g., using Crowdsourcing services, e.g., Amazon Mechanical Turk (AMT))
 - Or to correct the annotations produced by your initial system
 - Annotate something new that your system may benefit from finding (e.g., annotate subjunctive mood to improve translation)
 - Can we make annotation so much [fun](#) or so [worthwhile](#) so that it will be done for free?

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Part I:

Introduction to

Information Retrieval

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Aim and objectives

- What is information retrieval system and how it differs from database system?
- What are the different components of information retrieval system?
- What are the characteristics of text retrieval system?
- How to evaluate an information retrieval system?

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Information Retrieval - Definition

- Task Statement

Build a system that retrieves documents that users are likely to find relevant to their queries.
(Mostly unstructured documents)

→ Unstructured documents have seen a greater growth in the last decades (think of online postings, web-documents, social media, online forums, etc.)

[What is difference with structured databases?]

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Databases vs. IR

	Databases	IR
What we're retrieving	Structured data. Clear semantics based on a formal model.	Mostly unstructured. Free text with some metadata.
Queries we're posing	Formally (mathematically) defined queries. Unambiguous.	Vague, imprecise information needs (often expressed in natural language).
Results we get	Exact. Always correct in a formal sense.	Sometimes relevant, often not.
Interaction with system	One-shot queries.	Interaction is important.
Other issues	Concurrency, recovery, atomicity are all critical.	Issues downplayed.

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What about databases?

■ Examples of databases

- ☐ Banks storing account information
- ☐ Retailers storing inventories
- ☐ Universities storing student grades

Student Table

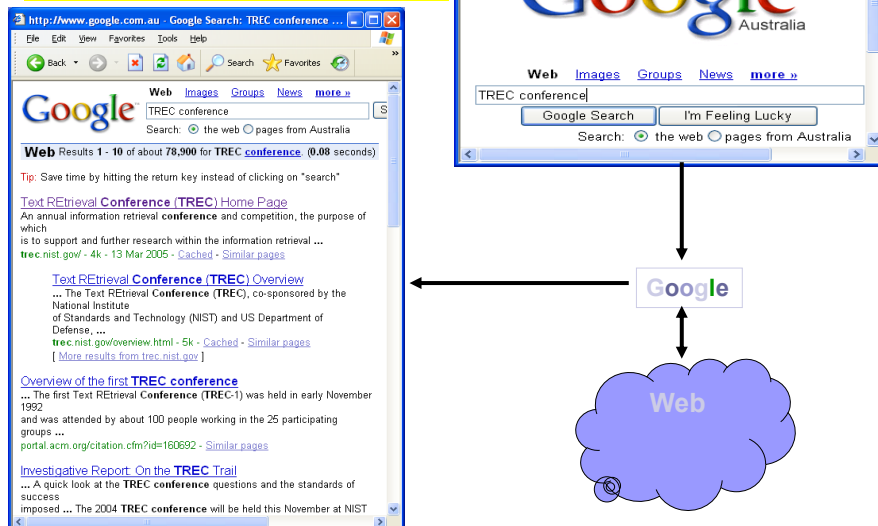
Student ID	Last Name	First Name	Department ID	email
1	Arrows	John	EE	jarrows@wam
2	Peters	Kathy	HIST	kpeters2@wam
3	Smith	Chris	HIST	smith2002@glue
4	Smith	John	CLIS	js03@wam

Course Table

Course ID	Course Name
lpsc690	Information Technology
ee750	Communication
hist405	American History

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Example of IR



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Information Retrieval Assumptions

- The system has available only pre-existing set of documents (text passages).
- Its response is limited to selecting from these passages and presenting them to the user.
- It must select, say, 10 or 20, or..., passages out of millions or billions!

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History of IR

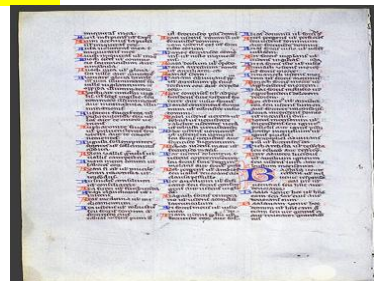
- First recent applications: **in libraries** (1950s)
 - ISBN: 0-201-12227-8
 - Author: Salton, Gerard
 - Title: Automatic text processing: the transformation, analysis, and retrieval of information by computer
 - Editor: Addison-Wesley
 - Date: 1989
 - Content: <Text>
- External attributes and internal attribute (content)
- Search by external attributes = Search in DB
- IR: search by content

<<<-- Key innovation
[How to make numbering effective?](#)

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History of IR (Cont.)

- Biblical Indexes and Concordances
 - 1247 – Hugo de St. Caro – employed 500 Monks to create keyword concordance to the Bible
- Journal Indexes (Royal Society, 1600's)
- Wilhelm Ostwald and Paul Otlet, 1910's: The "monographic principle" and Universal Classification. Used in worldwide libraries
- "Information Explosion" following WWII
 - Cranfield Studies of indexing languages and information retrieval --- basis of TREC conferences (<https://trec.nist.gov>)



Physical-chemistry



Universal decimal classification

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Retrieval task involves:

- Feature extraction: which feature to choose?
- Feature indexing
- Query formulation
- Feature matching
- Result visualization
- Feedback loop

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Match Task and Search Type (1)

- Search type varies with documents / databases employed.
- **WWW Tasks:** Find how the internet security works
 - Purchase plane tickets and hotel for
 - Find a trip to NY City.
 - Find a recipe for potato latkes.
- **Characteristics:**
 - Timely, specific, found via help from human agents and in well-known resources before the WWW.

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Match Task and Search Type (2)

- **Newsire & Magazine Tasks:** (from the TREC collection)
 - Find articles on research into cures for osteoporosis.
 - Find articles on the effects of recycling of tires on the environment.
 - Find information on prison overcrowding and how inmates are forced to cope with those conditions.
- **Characteristics:**
 - Complex combinations of topics.
 - Research-oriented
 - Either timely or retrospective

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How to handle search?

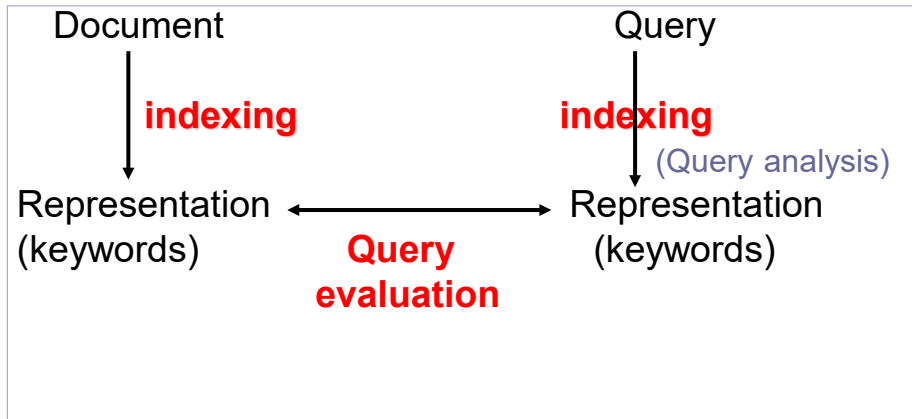
1. String matching (linear search in documents)
 - Slow
 - Difficult to improve
2. Indexing (*)
 - Fast
 - Flexible to further improvement

Remember: Indexing

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Indexing-based IR (Most used ones)



How easy to get the text indexed?

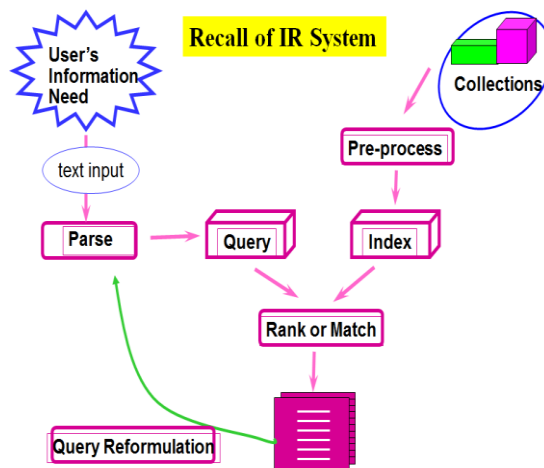
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User Interaction

Information retrieval meets computer-human interaction

How do we present search results to users in an effective manner?

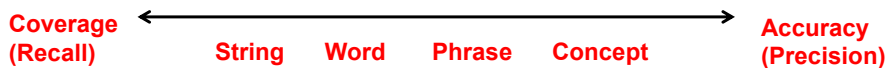
What tools can systems provide to aid the user in information seeking?



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What might be an Index?

- Goal = Find the important **meanings** and create an internal representation
- Factors to consider:
 - Accuracy to represent meanings (semantics)
 - Exhaustiveness (cover all the contents)
 - Facility for computer to manipulate
- What is the best representation of contents?
 - **Char. string** (char trigrams): not precise enough
 - **Word**: good coverage, not precise
 - **Phrase**: poor coverage, more precise
 - **Concept**: poor coverage, precise



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What might be an Index? Common practice

1. Use whole vocabulary of words extracted from the original set of documents
2. Use a restricted set of vocabulary (e.g., 1000 most frequent words excluding stopwords)
3. Use external vocabulary (e.g., Meriam dictionary; predefined list of named-entities)

→ See Lab sessions

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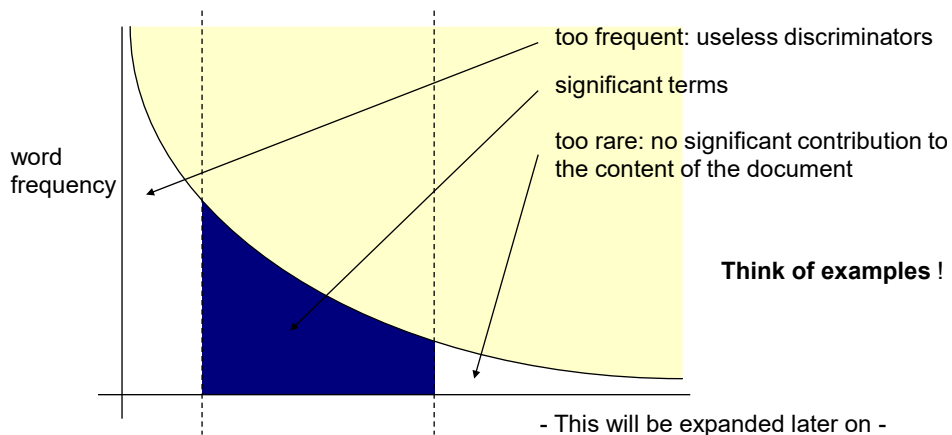
Why Text is Tough?

- Abstract concepts difficult to represent (AI-Complete)
- “Countless” combinations of subtle, abstract relationships among concepts
- Many ways to represent similar concepts
space ship, flying saucer, ...
- Concepts are difficult to visualize
- High dimensionality
Tens or hundreds of thousands of features

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Term significance

Word occurrence frequency is a measure for the significance of terms and their discriminatory power (see [Brown corpus](https://en.wikipedia.org/wiki/Brown_Corpus) (https://en.wikipedia.org/wiki/Brown_Corpus)).



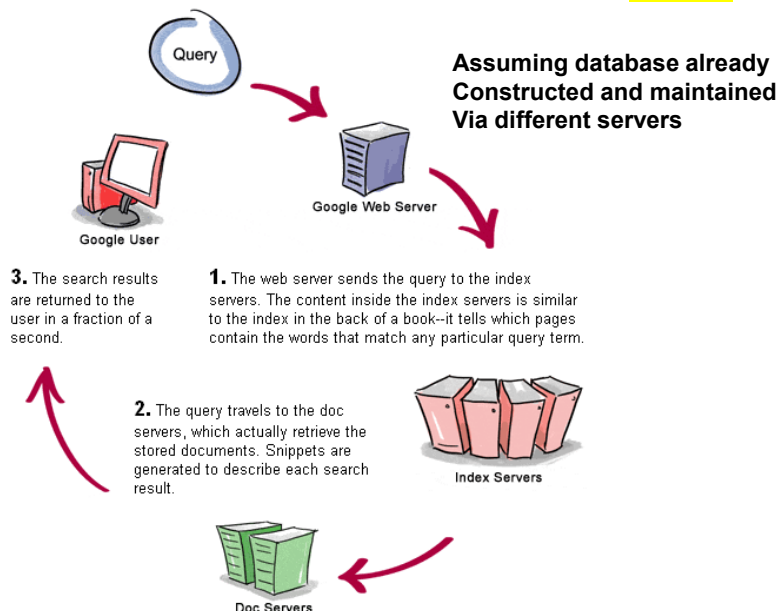
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How search engines work? Data Collection Process

- Autonomous “spiders”
 - Traversal Spider – “assembles lists of candidate Web documents that may include text document, images, videos, or hyperlinks to them”
 - Hyperlink Parser – “which extracts the Web addresses of doc, images and videos”
 - Content Spider – “which retrieves, analyzes, and iconifies the documents, images and videos”

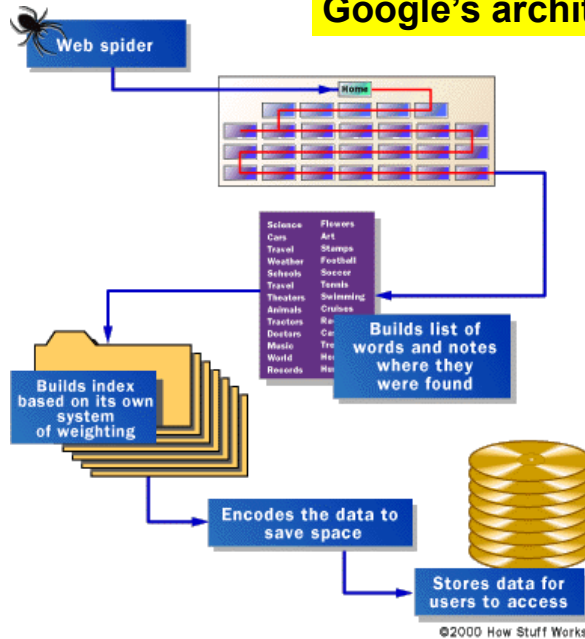
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How Google Works



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Google's architecture



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Sorting the search results: Page Rank in Google search engine

It is secret of search engine!

- Premise: importance of a research paper can be judged by the number of citations
- Analogy: importance of a Web page can be judged by the number of links pointing to it from other pages

Or, mathematically, where PR = page rank and C = citations (links) [original Google's formula]

$$PR(A) = \frac{(1-d)}{n+1} + d \left(\frac{PR(T1)}{C(T1)} + \dots + \frac{PR(Tn)}{C(Tn)} \right)$$

d is a damping factor in the range of $0 < d < 1$, usually 0.8

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Breaking Down the Equation

- **PR(Tn)** - Each page has a notion of its own self-importance. That's "PR(T1)" for the first page in the web all the way up to "PR(Tn)" for the last page
- **C(Tn)** - The citation count, or number of in-coming links, for page 1 is "C(T1)", "C(Tn)" for page n, and so on for all pages.
- **d(...)** - Damping factor multiplication restricts the influence of page rank score on total vote.
- **(1 - d)** - The $(1 - d)$ at the beginning ensures that the "*sum of all web pages' PageRanks will be one*": it adds in the bit lost by the **d(...)**. It also means that if a page has no links to it (no backlinks) even then it will still get a small PR of '1-d'.

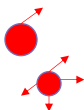
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Page Rank (continued)

The page rank of a Web page is the sum of the page ranks of all the pages linking to it divided by the number of links on each of those pages



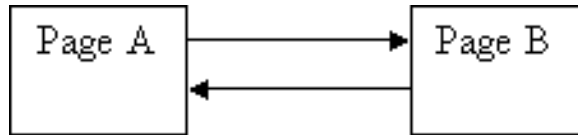
- A page with many links to it from other pages is judged to be more important than a page with only a few links from other pages



- A page with only a few links to other pages is deemed to be more important than a page with links to lots of other pages, e.g., a portal
- The damping factor allows the system to accommodate the user's view

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Simple Example

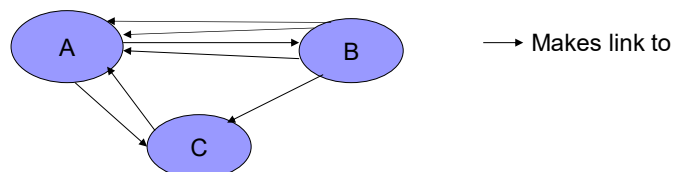


- Each page has one outgoing link. So that means $C(A) = 1$ and $C(B) = 1$.
- Assume $d=0.8$
- The page rank of A and B are given by solving eqs
$$P(A) = (1-d)/2 + d P(B)$$
$$P(B) = (1-d)/2 + d P(A)$$
Do it as an exercise!

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Tutorial

- Explain role of damping factor to account for human intervention
- Consider the 3 webpages A, B, C:



1. Determine the rank of each page as a function of damping Factor d
2. What is the rank of each page if damping $d=0.5$

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Evaluation of IR system

■ Quantitative evaluation

- **Precision**: proportion of true answers among all retrieved answers
 - **Recall**: proportion of true answers among all correct answers in document database
-
- Large test collections are needed to capture user variation, to support claims of statistical significance in results, to demonstrate that systems scale up ! Commercial credibility

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Evaluation (Cont.)

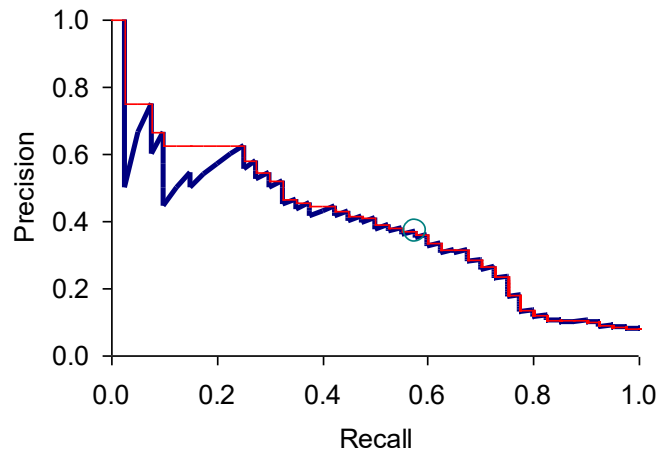
	Relevant	Nonrelevant
Retrieved	tp	fp
Not Retrieved	fn	tn

- Precision: $P = tp / (tp + fp)$
- Recall: $R = tp / (tp + fn)$
- **Accuracy**: the fraction of correct classification
 $(tp + tn) / (tp + fp + fn + tn)$
Not always a good measure for evaluation!
- **Recall** is a non-decreasing function of the number of docs retrieved

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Evaluation (Cont.)

A precision-recall curve example



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Difficulties in using precision/recall

- Should average over large document collection/query ensembles
- Need human relevance assessments
 - People aren't reliable assessors
- Assessments have to be binary
 - Nuanced assessments?
- Heavily skewed by collection/authorship
 - Results may not translate from one domain to another

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Standard relevance benchmarks

- TREC - National Institute of Standards and Technology (NIST) has run a large IR test bed for many years <http://trec.nist.gov/>
- Reuters and other benchmark doc collections, e.g., <http://www.daviddlewis.com/resources/testcollections/reuters21578/>
 - “Retrieval tasks” specified
 - sometimes as queries
 - Human experts mark, for each query and for each doc, Relevant or Nonrelevant
 - or at least for subset of docs that some system returned for that query

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Major Search Engines

- Google <http://www.google.com>
 - Well-deserved reputation as top choice for searching web
 - Crawler-based service provides comprehensive coverage and great relevancy
 - Recommended as a first stop
- Yahoo <http://www.yahoo.com>
 - Web's oldest (1994) directory
 - Shift in October 2002 to crawler-based listings from Google
 - Second shift in February 2004 to its own search technology
- Ask Jeeves <http://www.ask.com/>
 - Ask Jeeves initially gained fame in 1998 and 1999 as being the "natural language" search engine
 - Behind the scenes, editors monitored search logs and went out onto the web to located what seemed to be the best sites
 - Today, Ask Jeeves depends on crawler-based technology

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Meta Search Engines

Unlike search engines, metacrawlers don't crawl the web to build listings. Instead, they allow searches to be sent to several search engines all at once. The results are then blended together onto one page.

///Dogpile <http://www.dogpile.com>

- ❑ Searches a customizable list of search engines, directories and specialty search sites
- ❑ Displays results from each search engine individually

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Meta Search Engines (Cont.)

MetaCrawler

<http://www.metacrawler.com/>

- ❑ One of the oldest meta search services,
- ❑ MetaCrawler began in July 1995 at the University of Washington
- ❑ **An Internet multisearch engine (<http://www.metacrawler.com/index.html>); it searches nine other search engines (Lycos, Open Text, Alta Vista, Excite, WebCrawler, InfoSeek, Inktomi, Yahoo, and Galaxy).**

Name	Language
Dogpile	English
Excite	English
Info.com	English
Kayak.com	Multilingual
Mamma.com	
MetaCrawler	English
MetaGer	Multilingual
Mobisimo	Multilingual
Otalo.com	English
Publisher's clearinghouse Search and Win	
Searx	Multilingual
Skyscanner	Multilingual
Wego.com	Multilingual

List of current metasearch engines
From Wikipedia

Child-safe search engines

Like Google search but with extra filters to ensure safety of kids.

- Kiddle
- KidRex
- KidzSearch

Enterprise Search

Focus on enterprise assets

- Apache Solr
- ElasticSearch

Social Search

Focus on social aspect

- Ecosia
- ChaCha search
- Facebook
- Eurekster

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Name	Language	Alexa global ranking [hide]
AOL Search*	Multilingual	318 † ^[1]
Ask.com**	Multilingual	173 ^[2]
Baidu	Chinese	4 ^[3]
Bing	Multilingual	47 ^[4]
Dogpile***	English	10,603 ^[5]
DuckDuckGo*	Multilingual	214 ^[6]
Ecosia*	Multilingual	700 ^[7]
Exalead	Multilingual	116,560 ^[8]
Excite*	Multilingual	15,098 † ^[9]
Gigablast	English	135,070 ^[10]
Google	Multilingual	1 ^[11]
HotBot*	English	298,374 ^[12]
Lycos	Multilingual	36,396 ^[13]
MetaCrawler***	English	19,168 ^[14]
Mojeek	Multilingual	148,996 ^[15]
Qwant*	Multilingual	2,158 ^[16]
Searx***	Multilingual	93,323 ^[17]
Sogou	Chinese	162 ^[18]
Soso	Chinese	96 ^[19]
Startpage.com**	English	1,946 ^[20]
Swisscows*	Multilingual	232,931 ^[21]
WebCrawler*	English	4,264 ^[22]
Yahoo!*	Multilingual	7 † ^[23]
Yandex	Multilingual	1,060
Yippy***	English	137,607 ^[24]
Youdao	Chinese	260 ^[25]

* Powered by Bing ** Powered by Google *** Metasearch engine

Ranking of Search Engine By activities

Source:
Wikipedia: List of Search Engines

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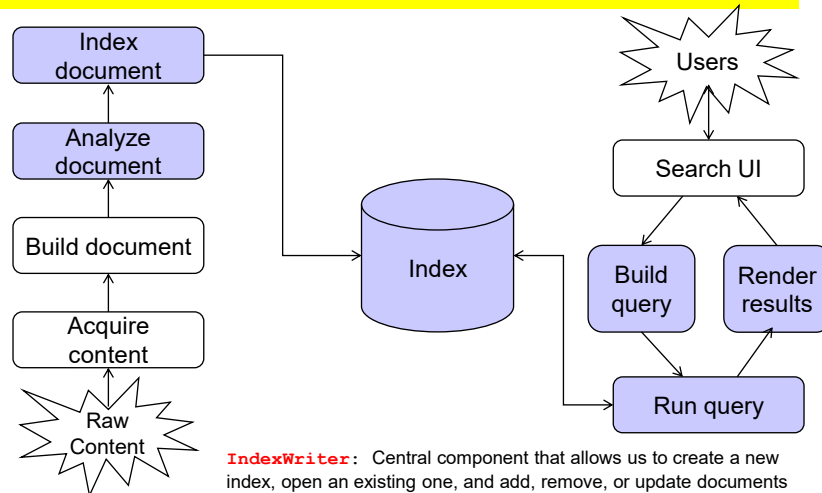


Lucene

- Open source Java library for indexing and searching
 - Lets you add search to your application
 - Not a complete search system by itself
 - Written by Doug Cutting
- Used by: Twitter, LinkedIn, Zappos, CiteSeer, Eclipse, ...
 - ... and many more (see <http://wiki.apache.org/lucene-java/PoweredBy>)
- Ports/integrations to other languages
 - C/C++, C#, Ruby, Perl, Python, PHP, ...
- Lucene in Action: <http://www.manning.com/hatcher3/>
 - Code samples available for download

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Lucene in a search system



IndexWriter: Central component that allows us to create a new index, open an existing one, and add, remove, or update documents in an index. Index encodes both freq and position information
Analyzer: Extract tokens from text Stream
Directory: Abstract class that provides location of index
 Lucene has tf-idf, BM25 similarity score methods implemented

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Other Resources

- Deep Web (Invisible Web)
 - Content of databases accessible on the Web
 - Estimated 500 times larger than the fixed Web
 - Non-textual files on the Web
 - Multimedia files
 - Graphical files (now offered by Google)
 - Sources
 - Directory of high quality databases maintained by Gary Price and Chris Sherman <http://www.invisible-web.net/>

Why website not found by search engine?

Restricted access, not added to open directory, no link to others (indexed) pages, inconvenient programming platform, etc.

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Search Engine -Challenges

- Search engines understand (and answer) questions in natural language
 - Microsoft Research's AskMSR
<https://www.microsoft.com/en-us/research/wp-content/uploads/2016/02/emnlp2002.pdf>
 - Rewrite search question (generate templates)
 - Search for exact matches – Not very effective
 - Search for close matches -- redundancy => high probability → but how to do it?
- Search engines as good as well-trained reference librarians
 - Major advances needed in probabilistic machine learning and natural-language processing
- **Developers → See: <http://www.google.com/apis/>**

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Conclusion

- Importance of information retrieval system in information technology and information access
- Information retrieval system provides guidelines to understand search engine mechanism.
- Search engine has 3 main steps: spider + index or catalog + retrieval page + display
- Challenges of future search engine: natural word (language) processing
- Perception and cognition should be accounted for

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<http://www-db.stanford.edu/~backrub/google.html>
- Google Search Engine <http://google.stanford.edu/>
- The original PageRank paper by Google's founders Sergey Brin and Lawrence Page
- <http://www-db.stanford.edu/~backrub/google.html>
- Good review of Gestalt psychology https://en.wikipedia.org/wiki/Gestalt_psychology

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Next

- Document representation and Models of Information Retrieval System.
- Introduction to Python toolkit for text analysis (NLTK) for labs. See online book
<http://www.nltk.org/book/>
- NLTK https://www.nltk.org/_library

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